

Assessing Credit Quality from the Equity Market: Can a Structural Approach Forecast Credit Ratings?

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Abstract

We investigate the empirical performance of default probability prediction based on Merton's (1974) structural credit risk model. More specifically, we study if distance-to-default is a sufficient statistic for the equity market information concerning the credit quality of the debt-issuing firm. We show that a simple reduced form model outperforms the Merton (1974) model for both in-sample fitting and out-of-sample predictability for credit ratings, and that both can be greatly improved by including the firm's equity value as an additional variable. Moreover, the empirical performance of this hybrid model is very similar to that of the simple reduced form model. As a result, we conclude that distant-to-default alone does not adequately capture the firm's credit quality information from the equity market. Copyright © 2007 ASAC. Published by John Wiley & Sons, Ltd.

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Résumé

Dans cet article, nous évaluons la performance empirique de la prédiction de probabilité de non-remboursement à partir du modèle de risque de crédit structurel de Merton (1974). Plus particulièrement, nous cherchons à examiner dans quelle mesure la "distance par rapport à la défaillance" (distance-to-default) est une statistique suffisante pour l'information du marché d'actions concernant la qualité du crédit de la compagnie émettrice. Nous démontrons qu'un modèle simple de formule réduite donne de meilleurs résultats à la fois pour l'agencement échantillonné et pour la prédictibilité hors-échantillon que le modèle de Merton (1974), et que ces deux résultats peuvent être encore plus précis si la valeur comptable de la firme est prise en compte comme variable supplémentaire. Par ailleurs, la performance empirique de ce modèle hybride est très semblable à celle du modèle simple de forme réduite. Aussi concluons-nous que la distance par rapport à la défaillance ne rend pas fidèlement compte de l'information concernant la qualité du crédit provenant du marché d'actions. Copyright © 2007 ASAC. Published by John Wiley & Sons, Ltd.

Mots-clés : Cote de crédit; probabilité de défaut; distance par rapport à la défaillance; modèle de risque de crédit structurel

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The proper assessment of credit risk has always been important to financial institutions. Recently, financial institutions have devoted even more resources to the development of tools that will adequately assess and manage the credit risk in their portfolios because, under the new Basel accord (Basel II) issued by the Basel Committee on Banking Supervision, regulatory capital will be determined using a bank's internal assessments of the probabilities that its counterparts will default. One popular approach to assessing credit risk involves the Merton (1974) model, where debts issued by a company are treated as derivatives written on the firm's underlying assets. This approach allows one to establish structural relationships among the firm's debt, equity, and the asset value. Using equity market data, one can generate a quantity that determines the default probability of a debt-issuing firm. These types of models, usually referred to as structural models, can thus establish relationships between default probabilities and some credit risk measures that are analytic functions of various relevant factors. Such models have become increasingly popular in recent years. In this paper, we investigate whether the credit risk measures inferred from structural credit risk models are adequate to reflect the credit risk information contained in those individual factors or, more generally, whether structural models can provide a better alternative to assessing firms' credit quality than those traditional statistical methods.

Empirical studies on the performance of structural models generally focus on the relationship between default risk and corporate bonds yields (see Eom, Helwege, & Huang, 2004 and the references therein). Few empirical studies have been conducted on the relationship between the actual default frequency and the theoretical default probability calculated from these models. Notable exceptions include Hillegeist, Keating, Cram, and Lundstedt (2004) and Vassalou and Xing (2004). Hillegeist et al. find that structural default probability measures contain relatively more information than Altman's Z-Score and Ohlson's O-Score. Using measures calculated from structural credit risk models as proxies to firms' default risks, Vassalou and Xing find that the Fama-French SMB and HML factors (see Fama & French, 1996) do not proxy the default factors. Moreover, in practice, Moody's KMV has been using the Merton (1974) type of structural models to predict the default probabilities of individual firms.

On the other hand, firms' credit qualities have traditionally been linked to their credit ratings, which are ordinal measures assigned by rating agencies to reflect the debt-issuing firms' ability to serve their debts. The rating process includes quantitative as well as qualitative and legal analysis. While similar approaches are used by

different rating agencies when assigning credit ratings, and although some agencies are more forthcoming than others in describing the procedures they follow in assigning or reviewing a rating, they all use proprietary methods to do so. For these reasons, various models have been proposed to predict a firm's credit rating changes based on publicly available information. Research in this area includes Blume, Lim, and MacKinlay (1998), Horrigan (1966), Kaplan, and Urwitz (1979), Pogue and Soldofsky (1969), and West (1973) among others. Despite the skepticism from rating agencies, these models have been fairly successful in explaining and predicting the ratings of a large cross-section of corporate bonds.

Most of the studies concerning credit rating forecasting have focused on either determining the factors affecting a firm's credit rating or building better econometric techniques. Popular approaches include hazard rate models, logistic type of models, and discriminant analysis, where usually the chosen independent variables enter the estimation process in a linear way. Little attention has been paid to the possible structural interactions among the independent variables and to the possible non-linear effect of the covariates.

Although credit ratings are not designed to pinpoint the precise probability of default, one should expect a close relationship between these two measures if one believes that credit ratings are sound measures of the credit qualities of the debt-issuing firms. Default probabilities derived from structural credit risk models, at least *prima facie*, provide a promising alternative for predicting credit rating changes. It is thus important to investigate whether these models can indeed perform better than the traditional statistical models in terms of in-sample fitting and out-of-sample prediction of credit rating changes. More specifically, we are interested in answering the following question: do the credit risk measures calculated from the structural models have better explanatory and predictive power for a firm's credit quality, which we assume are represented by the firm's credit rating, than the linear combination of those variables appearing in the structural models?

In spite of the importance of this subject, few studies have been conducted, with the exceptions of Geske and Delianedis (1998); KMV (1998); Leland (2002); and Vassalou and Xing (2003). Geske and Delianedis document that the default probability calculated from structural models contains enough information that both the rating migration and defaults are detected months before the actual credit events. Vassalou and Xing employ the default measure calculated from structural models and study the equity returns following changes in credit ratings. KMV compares the credit risk measure obtained from the credit transition matrix provided by rating

agencies with KMV's Expected Default Frequency (EDF) and finds that risk measurement systems based upon the rating transition matrix are subject to large errors. It reports that firms' expected default frequency with different ratings often overlaps. Leland (2002) compares the Leland and Toft (1996) model, which is based on an endogenously determined default boundary, with the Longstaff and Schwartz (1995) model, where default boundary is exogenously specified. He finds that the Leland and Toft model can generate a default probability term structure that fits better with the actual default probability term structure for a specific class of firms within the same credit rating.

None of these studies, however, address the issue of whether the default measures calculated from structural models are adequate in reflecting the credit information contained in the variables used as inputs to the models, such as the firm's equity value and its volatility and the debt-to-equity ratio, among others. Moreover, it is also worthwhile to investigate whether the default measures calculated from the structural models can actually predict a firm's credit risk, which we assume is reflected by the firm's credit rating, better than the ad hoc reduced form models. Alternatively, we investigated whether the default risk measures calculated from structural models adequately reflect all the information in the individual factors concerning the firm's credit quality. We attempted to answer these questions. We also compared the performance of these two types of approaches based on the performance of both their in-sample fittings and out-of-sample predictions.

Another motivation for our study originates from the well-documented empirical findings that structural credit risk models have difficulties in generating the large credit spreads observed in the corporate bond market, especially when the maturity of the bond is relatively short. Huang and Huang (2003) argue that credit risk contributes to only a fraction of credit spread (see also Ericsson & Renault, 2006). Instead, they attribute a major part of credit spread to the liquidity of corporate bonds. Their results are based on the assumption that the structural credit risk models used in their study are adequate to capture and price credit risk. As pointed out by Huang and Huang, their method creates a joint hypothesis problem: when a calculated credit yield spread based on a model fails to match the empirically observed yield spread, one may also conclude that the model is misspecified. In this paper, we shed some light on whether structural credit risk models can indeed adequately capture the default risks in the first place.

Our empirical results suggest that Merton's theoretical default measure, which is represented by *distance-to-default*, is not a sufficient statistic of the market

information concerning the credit quality of the issuing firm. Both in-sample fit and out-of-sample prediction can be greatly improved by using a hybrid model that includes the equity value of the firm as an additional independent variable, which suggests that the market information regarding the firm's credit quality is not fully captured through the single quantity of distance-to-default. Moreover, we find that the performance of this hybrid model differs little from a simple statistical model, where the main inputs for the structural model are used as explanatory variables. For this reason, we conclude that the credit risk measures calculated from the structural models are inadequate to capture the credit risk information contained in the variables for the model, and thus structural models do not necessarily provide any additional ability in capturing credit risks. A recent work by Bharath and Shumway (2004) reaches similar conclusions when historical default data are used in their analysis. On the other hand, a recent paper by Benos and Papanastasopoulos (2005) finds that by combining the risk-neutral distance-to-default measure and accounting factors one can predict credit rating changes fairly well.

The remainder of this paper is organized as follows: The next section introduces the structural credit risk models. Following this, we summarize how to compute the default probability measures for individual firms by using their equity prices. We then describe the econometric methodology used in this paper. Following this description, we present the data used and report the summary statistics. Before concluding we provide the empirical tests, discuss the estimation results, and investigate forecasting abilities of both structural and statistical models. The Appendix presents the estimation method for the hazard rate model.

Structural Credit Risk Model

In the seminal work of Merton (1974), the firm's asset value V_A is assumed to follow a lognormal process in the following form:

$$\frac{dV_{At}}{V_{At}} = \mu_A dt + \sigma_A dz_t, \quad (1)$$

where μ_A and σ_A are assumed to be constants that describe the instantaneous growth rate and volatility of the firm's asset value, respectively; and z_t is a standard Brownian motion. If we also assume that there is only one class of debt that pays no coupon and has a maturity time T and principal amount X , then the firm is in default if the firm's asset value is less than X at time T .

The equity of the firm, V_E , can be regarded as a call option written on the firm's underlying asset and can be

expressed through the Black and Scholes option pricing formula:

$$V_E = V_A N(d_1) - X e^{-rt} N(d_2), \quad (2)$$

where

$$d_1 = \frac{\ln(V_A/X) + (r + \sigma_A^2/2)T}{\sigma_A \sqrt{T}}, \quad d_2 = d_1 - \sigma_A \sqrt{T}, \quad (3)$$

r is the risk-free interest rate and $N(\cdot)$ is the cumulative probability function for standard normal distribution. The current value of the debt is given by $V_A - V_E$.

Models based on this approach are usually referred to as *structural models*. They include the following basic elements:

- Risk-free interest rate process: This specifies the dynamics of the interest rate without default risk. It is assumed to be constant in the Merton model (1974).
- Asset value process: The expected rate of return (including risk premium) and volatility of the asset value process reflect the expected business growth opportunities and the business risk, respectively.
- Face value of debt: The level of debt determines the leverage ratio of the firm, which is an important indicator of the issuing firm's credit quality.
- Default boundary: This specifies the asset level that causes the firm to default when its asset value declines to this level. In the Merton model, this boundary is characterized by the face value of the debt.
- Recovery rate: This postulates the value that the debt holders receive per \$1 of the value of the debt when the firm is indeed in default. This rate is not needed in the Merton model.

These components of structural credit risk models include all the variables considered important by rating agencies when assigning credit ratings. Structural models differ from traditional statistical ones in that the former specify how each component interacts with one another.

Various extensions of the Merton (1974) model have been proposed, including Collin-Dufresne and Goldstein (2001); Duffie and Lando (2001); Hull and White (1995), Leland (1998); Leland and Toft (1996); and Longstaff and Schwartz (1995), among others. Although the mathematical derivations of these models appear to be complex, the basic idea is simple; once we specify the stochastic process of a random variable, then theoretically we are able to calculate, either analytically or numerically, the probabilities of that variable being higher or lower than a prespecified boundary at any later point of time. For example, in the Merton model, the firm's asset value is assumed to follow a lognormal process, with the face value of the firm's debt being the default boundary. The probability that the firm's asset

value falls below the default boundary can be easily calculated analytically. In general, a structural credit risk model with a closed form solution for the corporate bond price has an analytical expression for the risk-neutral default probability, which is the theoretical measure of credit quality with structural explanation for the interaction of different risk factors and their effects on default. KMV Corporation adopted this approach when assessing default probabilities based on the Merton model. In closing this section, it is important to note that when deriving the physical default probability, what matters is the physical expected asset return rather than the risk-free interest rates used in the option pricing formula for the pricing purpose.

Estimation of Default Probability from Equity Data

If we assume that the firm's asset value follows the process given in (1), its debt amount of X matures at time T and the firm is in default if its asset value falls below level X at time T , then its default probability is given by:

$$P_{def} = P(V_{AT} \leq X),$$

where P is the physical or empirical probability. The firm's asset value at time T can be written as:

$$\ln(V_{AT}) = \ln(V_A) + (\mu_A - \sigma_A^2/2)T + \sigma_A z_T,$$

where V_A is the current value of the firm's asset. The firm's default probability can be calculated as:

$$\begin{aligned} P_{def} &= P(\ln(V_{AT}) \leq \ln(X)) \\ &= N\left(-\frac{\ln(V_A/X) + (\mu_A - \sigma_A^2/2)T}{\sigma_A \sqrt{T}}\right). \end{aligned}$$

This result shows that the distance-to-default, defined as:

$$Distance = \frac{\ln(V_A/X) + (\mu_A - \sigma_A^2/2)T}{\sigma_A \sqrt{T}}, \quad (4)$$

determines the firm's default probability.¹

Default happens when the ratio of the value of assets to debt is less than one. The distance-to-default measure indicates how many standard deviations the logarithm of this ratio needs to deviate from its mean before the firm defaults. In order to calculate this measure, we need to estimate the parameters μ_A and σ_A , which represent the firm's asset growth rate and volatility, respectively. However, since the asset value is not traded, these parameters cannot be directly observed from the market.

To compute the distance-to-default measure from equity values, we followed a similar procedure adopted

by KMV and Vassalou and Xing (2004).² We consider the default probability for a one-year period. All of the firm's debts were assumed due in one year, and the total amount of debt was approximated as the sum of all the short-term debts plus 50% of the long-term debts. The logic of this approach is that debts due within one year are more likely to cause default.

In order to calculate the asset value's growth rate μ_A and volatility σ_A , we first estimated the volatility of its equity σ_E as the standard deviation of its equity returns during the past twelve months. This estimation of σ_E is used as an initial value for the estimation of σ_A . Using this initial estimation of σ_A , we can obtain a time series of asset values V_A by inverting the equation (2) with market equity values and debt. The next step was to calculate the standard deviation of those values of V_A . This standard deviation will be used as the value of σ_A for the next round of iteration. This process is repeated until the standard deviations of V_A from two consecutive iterations converge.³ Once the monthly values of V_A are obtained, we can compute the growth rate μ_A as the mean return. Interest rates used in the calculation are the one-year constant maturity T-bill rates observed at the end of each month.

Alternatively, asset volatility and asset value can be calculated with the following equation:

$$\sigma_E = \frac{V_A}{V_E} \Delta \sigma_A, \quad (5)$$

where Δ is the hedge ratio $N(d_1)$ from equation (2). There are two unknown variables, V_A and σ_A , and two equations, (2) and (5), that are straightforward to solve.⁴ The drawback, however, is that equation (5) only holds instantaneously.

As pointed out by KMV (2002) and Vassalou and Xing (2004), the usual assumption that a firm's asset value follows a lognormal distribution may not be appropriate. For default measurement, the likelihood of large adverse changes in the relationship of the asset value to the firm's default point is critical to the accurate determination of the default probability. These changes may result from changes in asset value or changes in the firm's leverage. In fact, KMV first measures the distance-to-default as the number of standard deviations the asset value is from default. Then the distance-to-default values are mapped to historical default data to determine the corresponding default probability (see KMV for details).⁵

Econometric Methodology

Most of the corporate bankruptcy literature uses static, cross-sectional specification for default estima-

tion. However, Shumway (2001) shows that hazard rate models are more suitable for forecasting bankruptcy than the single-period models used previously in the literature. To overcome the difficulty in estimating hazard rate models with time-varying covariates, Shumway demonstrates that hazard rate models can be estimated from the simple logistic type of models plus panel data, called the *dynamic logistic model*. In order to estimate the hazard rate model in the case of credit ratings, which include the default state as a special case, Du (2003) extends Shumway's binomial single-cycle model to a multinomial and multicycle hazard rate model and shows that this model is also equivalent to a logistic type of model plus panel data. The proof of the equivalence is presented in the Appendix. We applied hazard rate models, or the dynamic logistic model, to conduct the statistical analysis.

Variables involved in computing the measure of distance-to-default included the following: risk-free interest rate r , firm's equity value V_A ,⁶ its volatility σ_E , firm's book value of debt X , and the growth rate of the firm's asset value μ_A . Thus, a naive statistical credit rating model employing equity market information takes the following form:

$$Z_t = \text{Intercept} + \beta_1 V_{E,t} + \beta_2 \sigma_{E,t} + \beta_3 \frac{V_{E,t}}{X} + \text{Dummy}_t + \varepsilon_t, \quad (6)$$

where Z_t is the latent credit quality score that jointly determines the probabilities of credit ratings and Dummy_t is the vector representing the year dummies variables. In addition, *Intercept* is a collection of cut-off points to separate the domain of the latent continuous variables Z_t into several intervals with each interval corresponding to a specific credit rating. Year dummies are included in the independent variables set to capture some time-varying effects, such as the changes in credit rating policies documented by Blume et al. (1998). When one takes the distance-to-default measure *Distance* as the only independent variable as suggested by the structural credit risk model in the third section, the competing logistic model is constructed as follows:

$$Z_t = \text{Intercept} + \beta \text{Distance}_t + \text{Dummy}_t + \varepsilon_t, \quad (7)$$

where the distance-to-default measure Distance_t is defined by equation (4).

Data and Summary Statistics

Each firm's current debt and long-term debt were obtained from the annual COMPUSTAT database. Following the approach adopted by KMV (2002) and Vassalou and Xing (2004), we used the "Debt in One Year"

plus half the “Long Term Debt” as the book value of a firm’s total debt. Monthly equity value data were also collected from COMPUSTAT. Although the amount of the long-term debt to be included is arbitrary because coupon payments of a firm are unobservable, this choice was sensible and adequately captures the financing constraints of a firm (see KMV). Furthermore, many firms do not use a calendar fiscal year. In order to match accounting data with credit ratings and provide a meaningful explanation for the year dummies, we chose calendar year data.

Credit ratings of each firm, or issuer ratings, rather than the ratings for particular debts are collected from the 2002 annual COMPUSTAT, which contains monthly data for corporate ratings from 1985 to 2002. Instead of using the finer credit ratings provided by Standard and Poor (S&P), we follow the conventions of the literature to classify all the ratings into the following eight categories: AAA, AA, A, BBB, BB, B, CCC, and CC. One advantage of employing issuer ratings instead of individual issue ratings is that the corporate rating will reflect the financial health and business risk of the firm rather than the bond-specific features. This also alleviates the concerns that *loss given default* also enters the decision process of bond rating agencies and helps to put more focus on the default probability of the firm represented by credit rating. Furthermore, cross-default clauses in debt contracts usually ensure that the default probabilities for each of the classes of debt for a firm are essentially the same; the default probability of the firm

determines the default probability for all the firm’s debt or counterpart obligations. Another advantage of using issuer credit ratings is that it greatly simplifies the computation of structural default risk measures. More specifically, option pricing models with multiple classes of debt are very complex and difficult to parameterize.

We collected monthly credit rating data for all firms with issuer credit ratings by S&P at the end of 1988. We used 1988 to compromise between two conflicting requirements: on one hand, more and more firms are rated by S&P over time, and in order to include more cross-sectional firms, it is desirable to begin with a later year.⁷ On the other hand, a longer time-series data set for each firm is preferred. In total, there were 1,508 firms in our sample. The individual firms in our sample were then tracked from 1989 to 2001.⁸ Following Vassalou and Xing (2004), monthly one-year constant maturity treasury bill rates were used for the one-year risk-free interest rates, and they were obtained from the Federal Reserve Board.

To calculate the distance-to-default measure for a firm, we applied the iteration method discussed in the third section. The estimated asset volatility converges very quickly. More than 60% of the total estimates took only two iterations to converge and most of them converged after seven iterations (93.5%). After we estimated the monthly asset values and asset value volatilities, we averaged the monthly values in each year to obtain the yearly values. In total, we had 6,635 firm year observations. Table 1 reports the summary statistics of the

Table 1
Summary Statistics of Independent Variables

Panel A Summary Statistics								
Variable	Mean	Median	Std.	Maximum	Minimum			
Asset/Debt	3.419	2.725	2.385	15.299	0.946			
Asset Volatility	0.064	0.055	0.042	0.461	0.018			
Equity/Debt	2.427	1.752	2.315	14.203	0.095			
Equity Volatility	0.100	0.084	0.062	0.403	0.028			
Asset Value	7.923	7.932	1.590	13.000	1.191			
Equity Value	7.390	7.446	1.773	12.842	0.394			
Growth Rate	0.006	0.006	0.018	0.062	−0.049			
Distance	6.103	5.614	4.016	31.528	−4.852			
Interest Rate	0.005	0.005	0.001	0.007	0.003			
Panel B: Means of Distance and Equity Value by Rating Category								
Variable	AAA	AA	A	BBB	BB	B	CCC	CC
Distance	10.742	9.828	7.679	5.513	3.317	2.348	1.444	0.638
Equity Value	10.322	8.711	8.034	7.446	6.237	5.097	4.284	3.209

calculated distance-to-default measure (distance), asset value, asset value volatility, asset value growth rate, as well as the asset-to-debt ratio (asset/debt) and equity value of the firm. We tracked the history for a sample of 1,508 firms with issuer credit ratings at the end of year 1988. The sample period covered 1989 to 2001. Each firm contributed multiple firm-year observations. A firm was deleted from the sample when its rating was withdrawn or if it went into default. In total, there were 6,635 observations with values for all the independent variables listed above, based on which summary statistics were calculated. Asset value, equity value, asset volatility, equity volatility, asset value growth rate, and distance were computed from monthly equity returns. Thus, the yearly average asset value volatility was approximately $0.064 \times \sqrt{12} = 0.222$. We also report the means of distance and equity value by rating category and default. To alleviate the effect of extreme values, we truncated all those variables to their 99th percentiles and 1st percentiles.⁹ To show the close relationship between the default risk measure suggested by the structural credit risk model, distance-to-default, and a firm's credit quality, we also report in Panel B the mean of the default measure for different credit categories. Panel B also contains the mean size of the firms' equity values in each of the credit categories. A glance at the table shows that the distance-to-default and equity value have a positive correlation with credit ratings. For example, the average of distance-to-default for rating AAA is 10.742, while that for rating BB is 3.317. This feature is consistent with the prediction from the structural models. Since distance-to-default is a function of equity value and asset value, as well as asset volatility, Table 2 reports the correlations among the independent variables used in this study. The results are consistent with the predictions from the structural model. Moreover, it also shows that none of the variables was redundant.

Table 2
Correlation Matrix of Elements of Distance-to-Default

	Distance	Equity Value	Asset Value	Asset Volatility
Distance	1			
Equity Value	0.42803	1		
Asset Value	0.32655	0.97742	1	
Asset Volatility	-0.46275	-0.09422	-0.1038	1

This table presents the correlation matrix of distance, equity value, asset value, and asset volatility. In total, there are 6,635 firm-year observations.

Results

In this section, we analyze relative performance between the different models discussed in the fourth section. Both in-sample fitting and out-of-sample predictability were investigated.

In-Sample Fitting Performance Comparisons

Table 3 Panel A reports the estimation results for the statistical credit rating model specified in equation (6). These estimates are for the dynamic logistic model of credit rating using a sample of 6,635 firm-year observations from 1989 through 2001. It models the probability of having higher ratings. Thus, the negative sign means that the variable has a negative effect on obtaining a higher credit rating. The equity volatility is calculated as the standard variance of equity returns in the last twelve months. We did not consider dividends. Intercepts divide the domain of latent continuous variable Z into several intervals. Each interval corresponds to each credit rating. Panel B reports the results for the model without the year dummies.

All three equity market driven covariates – equity-to-debt ratio, equity value, and equity return volatility – were statistically significant. In addition, the signs are consistent with the stylized empirical facts. For example, the higher equity value (positive coefficient) implies that it is less likely to default and thus has the larger probability of getting a higher credit rating. The adjusted R^2 of 0.6221 implies that the model can explain a major part of the variance of credit ratings. Furthermore, it is interesting to note that the year dummies display a steady downward trend over time, especially from 1990 through 1997, which is consistent with the empirical findings by Blume et al. (1998), who attribute this to the hypothesis that rating agencies have adopted a more stringent rating policy over the years. Panel B presents the same estimation when the year dummies variables are omitted from the independent variable set. The result is essentially the same as those in Panel A.

In order to show the empirical performance of the structural model, Table 4 Panel A presents the estimation results for the model specified in equation (7). The structural default probability measure, *Distance*, is statistically significant. Moreover, the positive sign of the coefficient is consistent with the structural model prediction: the larger the distance-to-default, the less likely the firm will default. However, all three model fitting measures – adjusted R^2 , AIC, and -2LogL – indicate that the model with the single independent variable *Distance* is not as good as traditional statistical models. Panel B presents the results when we omit the year dummy variable.

Table 3*Estimates of Statistical Credit Rating Model for the Panel Data, 1989–2001*

Parameter	Panel A				Panel B			
	Coef	Std. Err.	Chi Sqr.	P-value	Coef	Std. Err.	Chi Sqr.	P-value
Intercept1	−10.0646	0.1966	2619.987	<.0001	−9.8087	0.1825	2888.041	<.0001
Intercept2	−8.0049	0.1761	2067.187	<.0001	−7.7777	0.1609	2337.179	<.0001
Intercept3	−5.5415	0.1596	1206.202	<.0001	−5.3524	0.1440	1381.873	<.0001
Intercept4	−3.3158	0.1507	484.0935	<.0001	−3.1659	0.1350	549.8603	<.0001
Intercept5	−1.3764	0.1515	82.5773	<.0001	−1.2588	0.1357	86.0138	<.0001
Intercept6	2.9471	0.219	181.168	<.0001	2.9534	0.2021	213.5621	<.0001
Intercept7	5.4128	0.3161	293.261	<.0001	5.3843	0.3036	314.4644	<.0001
Equity Value	0.8151	0.0197	1714.453	<.0001	0.7791	0.0185	1764.703	<.0001
Equity/Debt	−15.1728	0.5173	860.2567	<.0001	−14.5622	0.4773	930.6691	<.0001
Equity/Debt	0.7979	0.0338	556.0896	<.0001	0.7728	0.0335	532.0852	<.0001
dummy1990	0.6196	0.1069	33.6177	<.0001				
dummy1991	0.3533	0.1079	10.7145	0.0011				
dummy1992	0.0843	0.1075	0.6145	0.4331				
dummy1993	−0.1326	0.1086	1.4915	0.2220				
dummy1994	−0.2259	0.1096	4.2466	0.0393				
dummy1995	−0.3077	0.1114	7.6286	0.0057				
dummy1996	−0.4074	0.1126	13.0817	0.0003				
dummy1997	−0.5212	0.1146	20.6882	<.0001				
dummy1998	−0.1209	0.119	1.0319	0.3097				
dummy1999	−0.00298	0.1222	0.0006	0.9805				
dummy2000	0.2143	0.1273	2.8352	0.0922				
dummy2001	−0.3066	0.128	5.7328	0.0167				
Adj- R^2	0.6221				0.6111			
AIC	15937.052				16085.999			
−2 Log L	15893.052				16065.999			

These estimates are for the dynamic logistic model (6) and Panel B reports the results for the model without the year dummies.

Although structural models suggest that a firm's credit quality is reflected in the distance-to-default measure, the poor fitting result suggests that this measure may not be adequate at reflecting the market information concerning the debt-issuing firm's credit quality. Results from Table 3 show that the firm's equity value is a significant factor affecting its credit quality besides its contribution to the equity-to-debt ratio.

To further illustrate this point, we investigated whether adding the equity value to the structural credit risk model could enhance the structural model's fitting ability. We constructed the following model:

$$Z_i = \text{Intercept} + \beta_1 \text{Distance}_i + \beta_2 V_{E,i} + \text{Dummy}_i + \varepsilon_i \quad (8)$$

We refer to this as the *mixed model*.

Table 5 Panel A provides estimation results for the mixed model specified in equation (8). In spite of its high correlation with the variable *Distance* as shown in Table

2, the firm's equity value is still statistically significant in explaining the firm's credit quality. The values of adjusted R^2 , AIC, and -2LogL show that the mixed model performs best among the three models we considered when fitting the historical data. This suggests that contrary to the theoretical results from structural models, the *Distance* measure itself is not sufficient in assessing a firm's credit quality from the equity market information. Panel B reports similar results for the mixed model (8) without the year dummy variable.

To see if the models can correctly predict a firm's credit rating, in Table 6 we show the percentage of correct rating predictions for each rating category. For brevity, we only compare the mixed model and the structural model. Specifically, we computed the predicted probability of a firm falling in each category, and the model-generated rating was assigned as the most likely rating predicted by the models, that is, the one with the

Table 4
Estimates of Structural Credit Rating Model for the Panel Data, 1989–2001

Parameter	Panel A				Panel B			
	Coef	Std. Err.	Chi Sqr.	P-value	Coef	Std. Err.	Chi Sqr.	P-value
Intercept1	−8.4143	0.1423	3496.929	<.0001	−7.452	0.118	3991.015	<.0001
Intercept2	−6.1994	0.1120	3065.673	<.0001	−5.3034	0.0826	4117.937	<.0001
Intercept3	−3.8425	0.0931	1703.975	<.0001	−3.0253	0.0589	2638.741	<.0001
Intercept4	−1.9686	0.0847	540.1750	<.0001	−1.2406	0.0494	630.2856	<.0001
Intercept5	−0.6405	0.0848	57.104	<.0001	0.00514	0.0525	0.0096	0.9221
Intercept6	1.8591	0.1292	206.955	<.0001	2.429	0.1123	468.0503	<.0001
Intercept7	3.8079	0.2872	175.838	<.0001	4.3677	0.2799	243.5546	<.0001
Distance	0.5187	0.0092	3174.28	<.0001	0.4552	0.0085	2870.048	<.0001
dummy1990	0.5258	0.1017	26.730	<.0001				
dummy1991	0.5725	0.1033	30.700	<.0001				
dummy1992	0.2396	0.1041	5.2987	0.0213				
dummy1993	−0.2104	0.1064	3.9130	0.0479				
dummy1994	0.2531	0.1075	5.5467	0.0185				
dummy1995	−0.0646	0.1090	0.3514	0.5533				
dummy1996	−0.0918	0.1089	0.7112	0.3990				
dummy1997	0.0476	0.1091	0.1909	0.6622				
dummy1998	0.8004	0.1113	51.682	<.0001				
dummy1999	1.3070	0.1147	129.804	<.0001				
dummy2000	1.5275	0.1186	165.738	<.0001				
dummy2001	1.1443	0.1197	91.3547	<.0001				
Adj- R^2	0.4734				0.4315			
AIC	17908.886				18359.856			
−2 Log L	17868.886				18343.856			

Panel A reports the results for the structural model in equation (7) and Panel B reports the results for the model without the year dummies.

highest predicted probability. With the minor exception of rating category BBB, the mixed model overwhelmingly dominated the structural models in every rating category. This again suggests that the distance-to-default measure alone is not a sufficient statistic in assessing a firm's credit quality.

On the other hand, the measure of distance-to-default is nevertheless one of the most important quantities that contain credit quality information of the debt-issuing firm. To illustrate its importance, we compared the in-sample fitting results for the following models that use various inputs for the structural model as covariates:¹⁰

- Equity value: $Z_t = \text{Intercept} + \beta V_{E,t} + \text{Dummy}_t + \varepsilon_t$;
- Volatility: $Z_t = \text{Intercept} + \beta \sigma_{E,t} + \text{Dummy}_t + \varepsilon_t$;
- Leverage ratio: $Z_t = \text{Intercept} + \beta V_{E,t}/X_t + \text{Dummy}_t + \varepsilon_t$.

The results are shown in Table 7. Although all variables are statistically significant, compared with the estimation

results in Table 4 Panel A for the structural model, values of adjusted R^2 , AIC, and -2LogL show that the variable *Distance* best describes the historical data. We also estimated the corresponding models without the year dummy variable. The results are similar and are thus not reported.

As we have shown, although *Distance* is an important quantity that contains a firm's credit quality information, it is not a sufficient statistic, and the in-sample fitting result can be greatly improved by adding the firm's equity value as an additional variable in the model. To test the robustness of this result, we also considered the following models:

- Model 1: $Z_t = \text{Intercept} + \beta_1 \text{Distance}_t + \beta_2 \sigma_{E,t} + \frac{V_{E,t}}{X_t} + \text{Dummy}_t + \varepsilon_t$;
- Model 2: $Z_t = \text{Intercept} + \beta_1 \text{Distance}_t + \beta_2 V_{E,t} + \beta_3 \sigma_{E,t} + \beta_4 \frac{V_{E,t}}{X_t} + \text{Dummy}_t + \varepsilon_t$.

Table 5
Estimates of Mixed Model for the Panel Data, 1989–2001

Parameter	Panel A			Panel B				
	Coef	Std. Err.	Parameter	Coef	Std. Err.	Parameter	Coef	Std. Err.
Intercept1	−14.3723	0.2133	4541.141	<.0001	−13.9654	0.2003	4863.245	<.0001
Intercept2	−11.8234	0.1835	4151.121	<.0001	−11.4758	0.1699	4564.554	<.0001
Intercept3	−9.1149	0.1601	3240.892	<.0001	−8.8405	0.1461	3660.549	<.0001
Intercept4	−6.7718	0.1427	2253.227	<.0001	−6.5569	0.128	2622.067	<.0001
Intercept5	−5.0048	0.1336	1403.62	<.0001	−4.8269	0.1185	1658.274	<.0001
Intercept6	−1.8888	0.1554	147.8241	<.0001	−1.7674	0.1429	153.0104	<.0001
Intercept7	0.2233	0.2900	0.5926	0.4414	0.3254	0.2877	1.2794	0.258
Distance	0.4091	0.0095	1869.38	<.0001	0.3797	0.00872	1897.15	<.0001
Equity Value	0.8603	0.0189	2077.369	<.0001	0.8247	0.0175	2216.421	<.0001
dummy1990	0.4352	0.1048	17.2579	<.0001				
dummy1991	0.3609	0.1065	11.472	0.0007				
dummy1992	0.0035	0.1073	0.0011	0.9738				
dummy1993	−0.5394	0.1096	24.2038	<.0001				
dummy1994	−0.2367	0.1110	4.5505	0.0329				
dummy1995	−0.5704	0.1127	25.6105	<.0001				
dummy1996	−0.7479	0.1130	43.827	<.0001				
dummy1997	−0.7837	0.1136	47.574	<.0001				
dummy1998	−0.3170	0.1165	7.4071	0.0065				
dummy1999	0.0404	0.1203	0.1126	0.7372				
dummy2000	0.1796	0.1246	2.0781	0.1494				
dummy2001	−0.2651	0.1262	4.4149	0.0356				
Adj- R^2	0.6391				0.6231			
AIC	15606.339				15843.098			
−2 Log L	15564.339				15825.098			

Panel A reports the results for the mixed model in equation (7) and Panel B reports the results for the model without the year dummies.

Table 6
Percentage of Correct Model-Generated Ratings

Rating	AAA	AA	A	BBB	BB	B	CCC + CC	Overall
Structural Model	9.14	18.51	52.94	64.17	7.50	26.95	0.00	40.94
Mixed Model	24.87	28.85	58.97	59.22	30.94	50.81	5.12	48.76

This table presents the percentage of correct model-generated ratings for the structural model and the mixed model. We compute the predicted probability of a firm falling in each category. The rating with the highest predicted probability is the model-generated rating.

The results are shown in Table 8.¹¹ In Model 1, the firm's equity value is replaced by its equity volatility and leverage ratio, while in Model 2 we included all the variables that are used in the structural mode including the firm's equity value, equity volatility, and leverage ratio. The results for Model 1 show that the variable *Distance* contains most of the credit information in the

equity volatility and leverage ratio, and the estimation results are hardly improved at all by including these variables as covariates. On the other hand, when the equity value is also included in the model, the performance of the model is almost the same as the mixed model. Moreover, certain collinearity will exist in this model, and its adjusted R^2 is slightly lower than that for the

Table 7
Comparison with Other One-Factor Models

Parameter	Equity Value		Volatility		Leverage Ratio	
	Coef	P-value	Coef	P-value	Coef	P-value
Intercept1	-12.1259	<.0001	-3.008	<.0001	-5.9571	<.0001
Intercept2	-10.0519	<.0001	-1.2908	<.0001	-3.8492	<.0001
Intercept3	-7.9765	<.0001	0.332	<.0001	-1.9238	<.0001
Intercept4	-6.237	<.0001	1.6676	<.0001	-0.4789	<.0001
Intercept5	-4.9049	<.0001	2.7197	<.0001	0.5801	<.0001
Intercept6	-2.3385	<.0001	5.1046	<.0001	2.8642	<.0001
Intercept7	-0.3659	0.218	7.158	<.0001	4.7794	<.0001
Variable	1.007	<.0001	-14.65	<.0001	0.4574	<.0001
dummy1990	0.0859	0.3936	0.1337	0.1751	0.2955	0.0029
dummy1991	0.0269	0.7931	0.2649	0.0082	0.2588	0.0104
dummy1992	-0.0435	0.6738	0.2276	0.0243	0.2024	0.0469
dummy1993	-0.1418	0.1774	0.2769	0.007	0.2354	0.023
dummy1994	-0.1922	0.0721	0.3572	0.0006	0.3551	0.0007
dummy1995	-0.2912	0.0071	0.2829	0.0073	0.3571	0.0008
dummy1996	-0.4441	<.0001	0.3168	0.0026	0.1927	0.0697
dummy1997	-0.619	<.0001	0.3817	0.0003	0.0276	0.7964
dummy1998	-0.696	<.0001	0.6607	<.0001	0.0852	0.4347
dummy1999	-0.8433	<.0001	0.7817	<.0001	0.1152	0.3017
dummy2000	-1.0311	<.0001	0.735	<.0001	0.1497	0.1918
dummy2001	-1.164	<.0001	0.6069	<.0001	0.0555	0.6336
Adj- R^2	0.4081		0.0894		0.2367	
AIC	18634.37		21350.04		20231.3	
-2 LogL	18594.37		21310.04		20191.3	

This table presents the estimation results for one-factor models in the following form:

$$Z_i = \text{Intercept} + \beta \text{Variable}_i + \text{Dummy}_i + \varepsilon_i,$$

where *Variable* can be equity value $V_{E,i}$, volatility $\sigma_{E,i}$, or leverage ratio $V_{E,i}/X_i$.

mixed model. It demonstrates that even after all the variables involved in the calculation of *Distance* have been included in the regression, the variable *Distance* is still significant. By comparing with the results in Table 3, we can see that this variable contains the nonlinear relationships between the debt-issuing firm's credit quality and the variables included in the statistical model. Moreover, the difference of the adjusted R^2 for Model 1 and Model 2 shows that the equity value of the firm is indeed an important factor for determining a firm's credit quality.

Out-of-Sample Prediction Performance Comparisons

We also investigated the out-of-sample predictability of the statistical model, the structural model, and the mixed model.¹² The percentage of correct credit rating predicting was used as an overall measure of their performances. We first divided our sample into two periods:

in-sample and out-of-sample. The observations from the in-sample period were used to estimate the parameters. Based on the estimated parameters together with the independent variables corresponding to the market information from the out-of-sample period, we used the models to predict the firm's credit rating corresponding to each out-of-sample data point. To show the consistency of the results, we used three different out-of-sample periods. The first column in Table 9 represents the time used to divide the whole sample into in-sample and out-of-sample periods. For example, year 1998 means that the in-sample observations include the period from 1998 through 1997, while the observations from 1998 to 2001 were classified into the prediction period. The percentage of correct predicting represents the overall predictability, regardless of specific credit ratings. Results in Table 9 show that the traditional statistical model systematically outperforms the structural credit rating model, while the mixed model gives the best out-of-sample predicting. We

Table 8
Comparison with Other Multivariable Models

Parameter	Model 1		Model 2	
	Coef	P-value	Coef	P-value
Intercept1	-8.4462	<.0001	-15.0944	<.0001
Intercept2	-6.1302	<.0001	-12.4802	<.0001
Intercept3	-3.724	<.0001	-9.7520	<.0001
Intercept4	-1.8303	<.0001	-7.4587	<.0001
Intercept5	-0.4954	<.0001	-5.7795	<.0001
Intercept6	2.0262	<.0001	-2.8805	<.0001
Intercept7	3.9931	<.0001	-0.8257	0.0095
Distance	0.4381	<.0001	0.4431	<.0001
Equity Value			0.8411	<.0001
Equity Volatility	-3.3686	<.0001	-0.3441	0.6438
Equity/Debt	0.1903	<.0001	0.0581	<.0001
dummy1990	0.5066	<.0001	0.4053	0.0001
dummy1991	0.5211	<.0001	0.3463	0.0011
dummy1992	0.194	0.0631	-0.0180	0.8664
dummy1993	-0.2088	0.05	-0.5581	<.0001
dummy1994	0.2389	0.0266	-0.2276	0.0398
dummy1995	-0.051	0.6409	-0.5808	<.0001
dummy1996	-0.1185	0.2779	-0.7475	<.0001
dummy1997	-0.0517	0.6371	-0.7587	<.0001
dummy1998	0.6353	<.0001	-0.2548	0.0286
dummy1999	1.1249	<.0001	0.1123	0.3514
dummy2000	1.3605	<.0001	0.2106	0.0915
dummy2001	0.9877	<.0001	-0.2225	0.0779
Adj- R^2	0.4889		0.6272	
AIC	17727.89		15805.43	
-2 LogL	17683.89		15759.43	

This table presents the estimation results for the following models:

Model 1: $Z_t = \text{Intercept} + \beta_1 \text{Distance}_t + \beta_2 \sigma_{E,t} + V_{E,t}/X_t + \text{Dummy}_t + \varepsilon_t$;

Model 2: $Z_t = \text{Intercept} + \beta_1 \text{Distance}_t + \beta_2 V_{E,t} + \beta_3 D_{E,t} + \beta_4 V_{E,t}/X_t + \text{Dummy}_t + \varepsilon_t$.

Table 9
Comparison of Forecasting Accuracy

Year	Structural Model	Statistical Model	Mixed Model
1998	38.45%	48.29%	50.55%
1997	39.76%	47.86%	48.61%
1996	39.66%	46.92%	48.07%

also conducted the same tests for each rating, and the results for 1996 are reported in Table 10 (the results for 1997 and 1998 show the same pattern and are not reported). The structural model under-predicts the credit quality of high-grade firms, and both the statistical model and the mixed model outperform the structural model significantly. The accuracy of the predictions for the statistical model and the mixed model are similar, at least for the firms with ratings between B and A.

Table 10
Percentage of Correct Model-Generated Ratings, 1996

Obs. Rating	Predicted Rating						
	Structural Model (1996)						
	AAA	AA	A	BBB	BB	B	CCC + CC
AAA	1.82	29.09	58.18	10.91	0	0	0
AA	1.44	5.26	55.02	36.36	0	1.91	0
A	0.45	3.14	37.72	52.99	1.2	4.49	0
BBB	0	0.54	16.85	65.9	2.17	14.54	0
BB	0	0	5.81	52.26	6.77	35.16	0.65
B	0	0	2.33	37.98	2.33	57.36	1.55
CCC + CC	0	0	0	6.67	6.67	86.67	13.33

	Statistical Model (1996)						
	AAA	AA	A	BBB	BB	B	CCC + CC
AAA	61.82	29.09	9.09	0	0	0	0
AA	16.27	45.45	25.84	11.96	0	0.48	0
A	3.14	15.72	57.78	22.01	0.3	1.05	0
BBB	0.27	3.26	45.38	44.16	4.21	2.72	0
BB	0	0.32	10.32	53.87	20.65	14.19	0
B	0	0	2.33	34.88	18.6	40.31	2.33
CCC + CC	0	0	0	26.67	6.67	40	13.33

	Mixed Model (1996)						
	AAA	AA	A	BBB	BB	B	CCC + CC
AAA	36.36	50.91	12.73	0	0	0	0
AA	4.78	41.63	38.76	14.35	0.48	0	0
A	0.3	11.83	60.78	26.5	0.6	0	0
BBB	0.14	1.9	37.77	52.58	6.93	0.68	0
BB	0	0	8.39	55.16	24.52	11.94	0
B	0	0	3.1	29.46	24.03	42.64	0.78
CCC + CC	0	0	0	0	26.67	66.67	6.67

This table presents the percentage of correct model-generated ratings. We compute the predicted probability of a firm falling in each category. The rating with the highest predicted probability is the model-generated rating.

The logistic models rely on a particular parametric functional form for the relationship between the credit ratings and the independent variables. One may argue that the credit risk measure distance-to-default is mapped to the real default data in a nonparametric way and is therefore not especially sensitive, at least on average, to model mis-specification. To capture this potential non-linear effect, we investigated the hazard rate model with third-order polynomials¹³ of distance-to-default:

$$Z_i = \text{Intercept} + \beta_1 \text{Distance}_i + \beta_2 \text{Distance}_i^2 + \beta_3 \text{Distance}_i^3 + \varepsilon_i \quad (9)$$

Although, in some cases, the second and third order of distance-to-default are statistically significant, our unreported empirical results show that the out-of-sample prediction accuracy is actually reduced due to overfitting compared with the original structural models.

Concluding Remarks

We empirically investigated whether structural approach is a better alternative for assessing credit quality using equity market information. Specifically, we

investigated whether the credit risk measure from the structural models is adequate at reflecting the market information contained in the variables in the model. Both in-sample fitting and out-of-sample prediction showed that the structural credit risk model does not generate a sufficient statistic of the market information concerning credit quality. For example, we found that the empirical performance of in-sample fitting and out-of-sample prediction of the model can be greatly improved by adding the equity value to the model. In other words, credit quality information contained in the equity value of the firm is not fully utilized by structural credit risk models. For this reason, it seems that structural credit risk models, although theoretically appealing, should be used with caution when assessing a firm's credit quality.

Although our results showed that the structural model has difficulty in capturing credit risk, they provide indirect support to the conclusion of Huang and Huang (2003) that credit risk accounts for only a fraction of the observed credit spread, especially for investment grade bonds. Specifically, by missing equity value, which is negatively correlated with default probability, they obtained a default probability measure biased upward for investment-grade firms whose equity size is large. Hence, the model credit spreads would be larger than what can be justified by the actual default probabilities. Yet even this larger model spread is only a fraction of the actual spreads observed in the market. On the contrary, our results suggest that structural models are not able to sufficiently capture and price credit risk. Our results are also consistent with the findings of Eom et al. (2004), who systematically examine five structural credit risk models¹⁴ and find that all of them have difficulties in accurately forecasting credit spreads.

Implications for Theory and Practice

Our results are relevant to both researchers and practitioners in finance. Recent studies concerning the credit risk, such as Vassalou and Xing (2003, 2004), often used the measure of distance-to-default derived from the Merton model as a proxy for the debt-issuing firm's credit quality. Moreover, the newly adopted Basel II specifically requires financial institutions to implement models to adequately measure the default risk of their portfolios. Our results suggest that the distance-to-default measure from the Merton model should be used with caution because it does not adequately reflect all the credit quality information contained in the variables involved in constructing the measure.

Limitations and Directions for Future Research

Finally, our conclusions are based on the assumption that credit rating is an accurate and timely measure of credit quality. It is, however, possible that the Merton type of models do reflect the *actual* default probability of each firm, yet the credit ratings assigned by rating agencies do not sufficiently reflect a firm's credit quality¹⁵ or credit ratings do not reflect a firm's financial health in a timely way. As a result, KMV's measure is not able to predict the credit rating very well. Moreover, we only studied the Merton model, and it would be interesting to see whether other more complicated structural models, such as Leland and Toft (1996) and Collins-Dufresne and Goldstein (2001), can adequately reflect the credit default risk of the debt-issuing firm.

Notes

- 1 The drift term $\mu_A - \sigma_A^2/2$ for the logarithm of the asset value is usually ignored in the definition of distance-to-default. However, the resulting value will be very close since the model is often used for a one-year time horizon, and the drift term is very small.
- 2 There are two main differences between Vassalou and Xing (2004) and KMV. KMV uses a more complicated method to assess the asset volatility than Vassalou and Xing, which incorporates Bayesian adjustments for the country, industry, and size of the firm. KMV also allows for convertibles and preferred stocks in the capital structure of the firm, whereas Vassalou and Xing allow only equity, as well as short- and long-term debt. In this study, we follow the Vassalou and Xing approach.
- 3 Our tolerance level for convergence is $2 \times 10E - 4$.
- 4 This approach was originally proposed by Ronn and Verma (1986). The parameters in the structural model can also be estimated by using the maximum likelihood estimation; see Duan (1994) and Ericsson and Reneby (2005) for discussions.
- 5 KMV has a proprietary default database including more than 250,000 firm-years observations and more than 4,700 incidents of default or bankruptcy. It has a procedure to map the distance-to-default to the empirical data and calculate the expected default frequency.
- 6 Following conventions of the literature, we take the logarithm of the equity value.
- 7 Because of the aging effect documented by Altman (1989), we do not include newly rated firms after 1988 into our sample (i.e., newly rated firms and seasoned firms have different rating transition probability).
- 8 Some firms may drop from our sample before 2001, either because of default or because of their ratings being withdrawn. For example, if a firm defaulted in, say, 1993, then that firm will be deleted from the sample since 1993. In other words, only the observations of that firm over the period of 1988 to 1992 will be included in the sample.
- 9 We truncate the estimated asset value volatility at the end of each iteration.

- 10 We thank the referee for this suggestion.
- 11 To save space, we omit the standard errors and Chi-square values of the estimations.
- 12 We have also considered the forecasting performance for the model in equation (9). The results are very similar to the case of the mixed model.
- 13 The authors thank Frank Zhang of Moody's KMV for this suggestion.
- 14 The models studied include Merton (1974), Leland and Toft (1996), Longstaff and Schwartz (1995), and Collin-Dufresne and Goldstein (2001).
- 15 The authors thank Professors Darrell Duffie and Frank Milne for their comments on this point.
- 16 A cycle begins when the firm enters a new credit rating and ends when it migrates to another rating

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Appendix

Estimating Hazard Rate Models

We assume that credit rating migration can happen at any of the following discrete times, $t = t_0 + \Delta s$, $t_0 + 2\Delta s$, $\dots$, where Δs can be very small to approximate the continuous time setting. We use T_i to represent the time when firm i leaves our sample (for ease of notation, we omit the subscript i).

Let the probability of transition from credit rating l to rating m at time $t + s$ be given by $p_{lm}(t, x, s)$, where s measures the duration that rating l has been continuously occupied since it was entered at time t and x represents the independent variables, or the covariates, which can be time-varying.

Let $\theta_{lm}(t, x, s)$ represent the transition intensity from rating l to rating m , given that state l was entered at time t and has been occupied up to time $t + s$. The variable $\theta_{lm}(t, x, s)\Delta s$ represents the probability at time t that a departure from rating l to state m occurs during the time interval from $t + s$ to $t + s + \Delta s$. This probability is conditional on the rating l being occupied for s period of time since t , and upon the previous transition history. Furthermore, we let c denote the c 'th cycle¹⁶ of the rating migrations process, with the total number of cycles being C . In addition, we let S_c be the duration of the c 'th cycle. We assume that rating migration only occurs at the end of each cycle. We also define a binary indicator d_{cm} , which equals 1 if rating m is entered at the end of the c 'th cycle, and 0 otherwise.

Proof of the Equivalence

Shumway's (2001) model is the simplest hazard rate model. Specifically, there are only two states in his model: default and not default. Furthermore, if a firm defaults, it will be removed from the data sample only after the default occurs. Shumway's (2001) model is a binary, single-cycle hazard rate model. This model is sufficient for studying bankruptcy events. For credit rating, however, each firm has multiple possible destinations when it experiences a rating change. For example, its rating can transfer from AAA to AA, or BB, or CC, or even default. In addition, each firm can experience multiple rating migrations during a particular period of time. In this Appendix, we show that Shumway's (2001)

approach can be extended to the multinomial, multiperiod case and demonstrate the equivalence of the hazard rate model and the dynamic logistic model. Equivalence refers to their likelihood functions and estimates being identical.

Proposition: A multiperiod, multinomial logistic model is equivalent to a hazard rate model with hazard function $P_{lm}(t, x, s)/\Delta s$.

Proof: A multiperiod, multinomial logistic model is estimated with each panel data point, say each firm-year, as if it were a separate observation. The contribution of a specific firm i to the likelihood function during its cycle in the sample is given as follows:

$$l = \prod_{c=1}^C \prod_{l=1}^K \left(\prod_{s_c=0}^{S_c} P_{ll}(t_c, x, S_c) \right)^{d_{c-l}} \prod_{m=1}^K P_{lm}(t_c, x, S_c)^{d_{cm}d_{c-l}}; \quad l \neq m, \quad (10)$$

where S_c is the completed duration of cycle c and t_c is the calendar time at which the c 'th cycle was commenced. Credit rating changes only happen at the end of each cycle after a rating was entered.

Equation (10) is intuitive. In its life cycle, a firm could transit within all possible credit rating states, including the bankrupt state, or stay in its original rating. The likelihood contributed by any specific firm is the product of the transition probabilities and/or the probability of staying in its original rating. For example, suppose a firm's rating at time t_0 was AAA, then time $t_0 + \Delta s$ was downgraded to AA, then time $t_0 + 2\Delta s$ downgraded to B, where it remained. The likelihood function contributed by this firm is $P_{AAA,AA}(\cdot)P_{AA,B}(\cdot)P_{B,B}(\cdot)$. An implicit assumption made here is that the latent variable solely determines the credit rating migration probability.

Because Δs is a constant, we can divide equation (10) by polynomials of Δs without changing the maximum likelihood estimation results. Then we have:

$$l' = \prod_{c=1}^C \prod_{l=1}^K \left(\prod_{s_c=0}^{S_c} P_{ll}(t_c, x, S_c) \right)^{d_{c-l}} \prod_{m=1}^K \left[\frac{P_{lm}(t_c, x, S_c)}{\Delta s} \right]^{d_{cm}d_{c-l}}; \quad l \neq m. \quad (11)$$

We get the following log likelihood function:

$$L = \sum_{c=1}^C \sum_{l=1}^K \left(\sum_{m=1}^K d_{c-l} d_{cm} \log \left[\frac{P_{lm}(t_c, x, S_c)}{\Delta s} \right] + d_{c-l} \log \prod_{s_c=0}^{S_c} P_{ll}(t_c, x, S_c) \right), l \neq m. \quad (12)$$

Let $P_{lm}(t_c, x, s_c) = \theta_{lm}(t_c, x, s_c) \Delta s$. Recall that for $l \neq m$ (see Gouriéroux (2000)),

$$\prod_{s_c=0}^{S_c} P_{ll}(t_c, x, s_c) \approx \exp \left[- \int_0^{S_c} \sum_{m=1}^K \theta_{lm}(t_c, x, u) du \right]. \quad (13)$$

Then we arrive:

$$L = \sum_{c=1}^C \sum_{l=1}^K \left(\sum_{m=1}^K d_{c-l,l} d_{c,m} \log \theta_{lm}(t_c, x, s_c) - d_{c-l,l} \int_0^{S_c} \sum_{m=1}^K \theta_{lm}(t_c, x, u) du \right), l \neq m. \quad (14)$$

Rearranging the terms, we have:

$$L = \sum_{c=1}^C \sum_{l=1}^K \sum_{m=1}^K d_{c-l,l} [d_{c,m} \log \theta_{lm}(t_c, x, s_c) - z_{lm}(t_c, x, s_c)], \quad (15)$$

where

$$z_{lm}(t_c, x, s_c) = \int_0^{S_c} \sum_{m=1}^K \theta_{lm}(t_c, x, u) du.$$

Equation (15) is exactly the likelihood function for the multiperiod, multinomial hazard rate model discussed

in Lancaster (1990). Therefore, if we let the probability density function of transition from credit rating l to rating m be given by $P_{lm}/\Delta s$, the likelihood function of a multiperiod logistic model is equivalent to the likelihood function of a discrete-time hazard rate model.

Inference of Hazard Rate Models

Making statistical inferences in a hazard model estimated with a logistic program is straightforward. Since the dynamic logistic and hazard models have the same likelihood function, they have the same asymptotic covariance matrix. The χ^2 test statistics provided by the logistic program take the following form:

$$(\hat{\mu}_k - \mu_0)' \Sigma^{-1} (\hat{\mu}_k - \mu_0) \sim \chi^2(k), \quad (16)$$

where there are k estimated moments being tested against k null hypotheses μ_0 and Σ is the estimated covariance matrix.